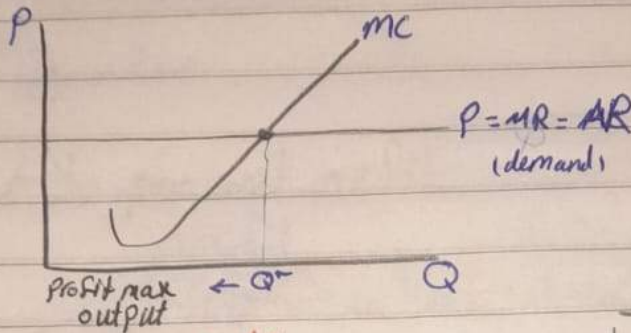


\* zero profits in long run

\* possible profit in short run

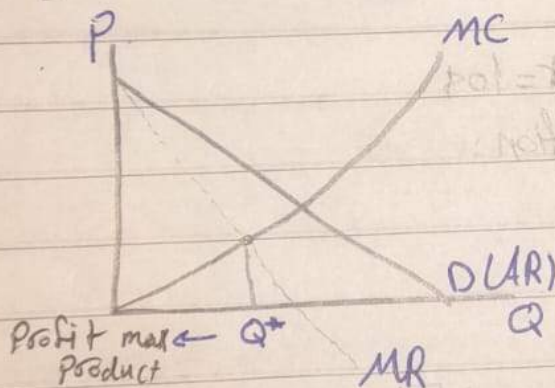
perfect competition

large number of firms (producers)



(price takers)

monopoly  
only one firm (producer)



(price maker)

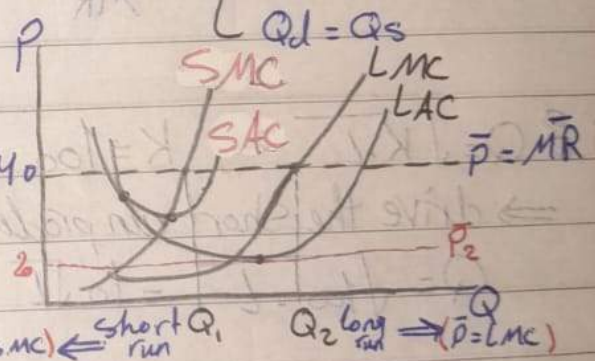
\* Profit maximization condition:

$$MC = MR$$

conditions of perfect competition:

\* in short run  $\rightarrow P = MR = MC$

\* in long run  $\left\{ \begin{array}{l} P = MR = MC \\ P = AC: \text{zero profit} \end{array} \right.$

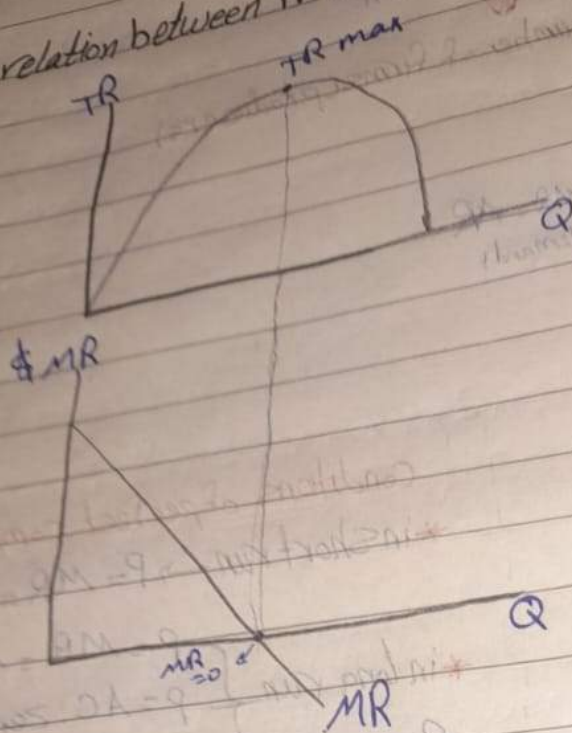


\* in long run with perfect competition

$\Rightarrow$  Freedom of entry

\* New firms will enter the market leading to zero profit

⇒ The relation between TR, MR



②  $Q = \sqrt{K \cdot L}$ ,  $K = 100$ ,  $w = 5\$$ ,  $r = 10\$$   
 ⇒ drive the short run production function:

$$Q = \sqrt{100 \cdot L} = 10\sqrt{L}$$

⇒ TC, MC function:

$$TC = w \cdot L + r \cdot K$$

$$TC = 5 \cdot L + 10 \cdot K^{(100)}$$

$$TC = 5L + 1000$$

$$TC = 5 \frac{Q^2}{100} + 1000$$

$$TC = 1000 + \frac{1}{20} Q^2$$

$$MC = \frac{\Delta TC}{\Delta Q} = \frac{\sigma TC}{\sigma Q}$$

$$MC = \frac{1}{10} Q$$

$$Q = 10\sqrt{L}$$

$$Q^2 = 100L$$

$$L = \frac{Q^2}{100}$$

if the firm is a price taker,  $P=20$  Find the profit maximization output?

$$MC = MR \rightarrow P$$

$$MC = 20$$

$$\frac{1}{10}Q = 20$$

$$Q = \frac{20}{\frac{1}{10}} = 200$$

Calculate the per unit profit:

$$\frac{\text{profit}}{Q}$$

$$\text{profit}(\pi) = TR - TC$$

$$P \cdot Q \quad 1000 + \frac{1}{20}Q^2$$

$$(20 \times 200) - (1000 + \frac{1}{20} \times 200^2)$$

$$\pi = 4000 - 3000 = 1000$$

$$\text{per unit profit} = \frac{1000}{200} = 5 \$/\text{unit}$$

Perfect Competitive Firm!

(ex)  $TC = 100 + Q^2 + Q$ , market (D)  $\Rightarrow P = 1000 - 2Q$ , market (S)  $\Rightarrow P = 100 + Q$

\* Find equil  $P, Q$ ?

$$1000 - 2Q = 100 + Q$$

$$3Q = 900$$

$$Q = 300$$

$$P = 1000 - 2 \times 300$$

$$P = 400$$

\* Find the profit maximization output of this firm? , C, profit

|                                                                                                                      |                                                      |                                     |                                                             |
|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|-------------------------------------|-------------------------------------------------------------|
| $p = 400 \rightarrow MR$<br>$MR = MC$<br>$400 = MC$<br>$400 = 2Q + 1$<br>$2Q = 400 - 1$<br>$2Q = 399$<br>$Q = 199.5$ | $P \times Q$<br>$400 \times 199.5$<br>$= 79800 = TR$ | $TC = 100 + Q^2 + Q$<br>$= 4099.75$ | $\pi = TR - TC$<br>$\pi = 79800 - 4099.75$<br>$\pi = 39700$ |
|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|-------------------------------------|-------------------------------------------------------------|

Show if this situation refers to short or long run?

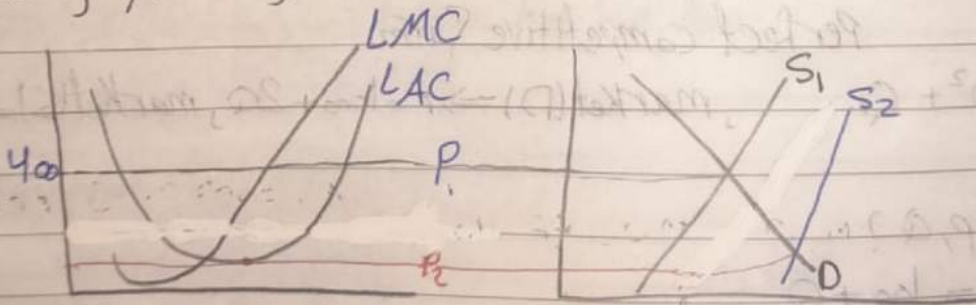
Short run  $\Rightarrow \pi \rightarrow$  positive.

long run  $\Rightarrow \pi = 0$

\* What do you anticipate to this firm at the long run?

Freedom of entry and exit will generate incentives for new entities to enter is market which leads to lower price. Thus zero profit.

\* Show graphically:



Suppose a perfect competitive firm;  $LAC: 40 - Q + 0,01Q^2$ ,  $LMC = 40 - 2Q + 0,03Q^2$   
 market (D):  $Q = 25000 - 1000P$

→ Find the long run equilibrium output?

$$P = MR = MC \text{ "max profit"}$$

$$P = AC \text{ "zero profit"}$$

$$Q_d = Q_s \quad (Q_d = Q^* \times n)$$

number  
of firms

$$P = 40 - 2Q + 0,03Q^2$$

$$P = 15$$

$$\therefore P = MC$$

$$P = 40 - 2Q + 0,03Q^2 \rightarrow 0$$

$$P = AC$$

$$P = 40 - Q + 0,01Q^2 \rightarrow 0 = \pi$$

①, ②

$$40 - 2Q + 0,03Q^2 = 40 - Q + 0,01Q^2$$

$$\frac{Q}{Q} = 0,02 \frac{Q}{Q}$$

$$0,02Q = 1$$

$$Q = \frac{1}{0,02} = 50$$

→ Find number of firms

$$Q_d = Q^* \times n$$

$$n = \frac{\text{demand}}{Q} = \frac{25000 - 1000 \times 15}{50} = 200$$

Ex) if a monopolist produce at a constant AC, MC,  $AC = MC = 5$ , he faces a market (d)  $\Rightarrow Q = 55 - P$ ,  $MR = 55 - 2Q$

\* Calculate the profit max Q, P,  $\pi$

$$MR = MC$$

$$55 - 2Q = 5$$

$$55 - 5 = 2Q$$

$$Q^* = 25$$

$$25 = 55 - P^*$$

$$P^* = 30$$

$$\pi = TR - TC$$

$$P \times Q - AC \times Q$$

$$30 \times 25 - 5 \times 25 = 625$$

\* if perfect competition find  $Q^*$   
 $(MR=P)=MC$   
 $p^*=5$

$$Q=55-p$$

$$Q=55-5=50$$

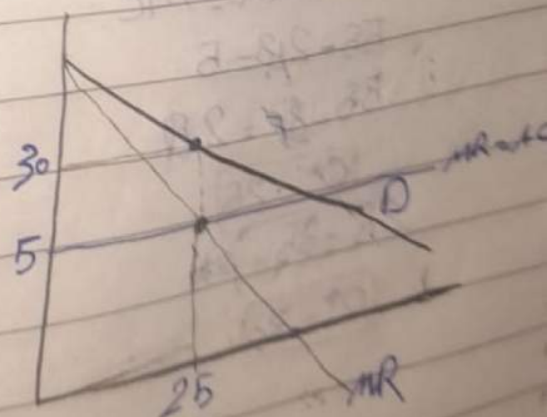
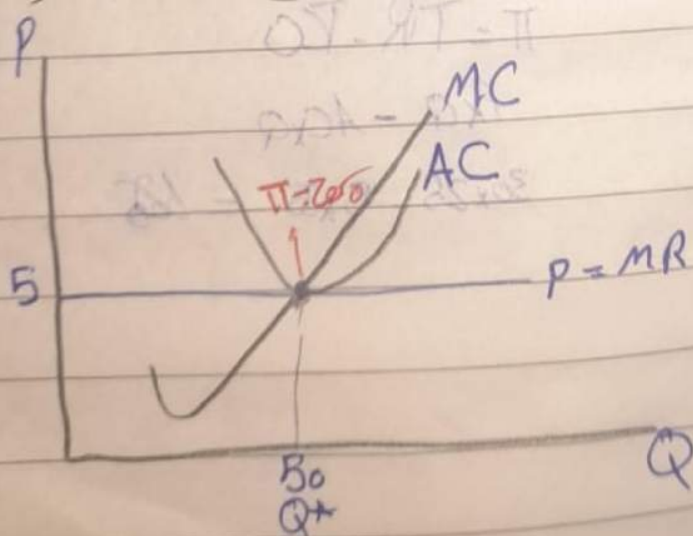
\* Profit if perfect competition  
 $\pi = TR - TC$   
 $250 - 250 = \text{Zero}$

\* Compare between 2 situations.

| monopoly  | Perfect competition |
|-----------|---------------------|
| $Q=25$    | $Q=50$              |
| $p=30$    | $p=5$               |
| $\pi=625$ | $\pi=\text{Zero}$   |

monopolist has a higher price and lower quantity but at Perfect competition lower price, higher quantity.

Show graphically.



(ex) A firm has the following D function  $\Rightarrow Q = 90 - 2P$

\* Calculate TR in terms of Q

$$P \times Q$$

$$Q = 90 - 2P \Rightarrow 2P = 90 - Q$$

$$P = 45 - 0.5Q$$

$$TR = P \times Q$$

$$TR = 45Q - 0.5Q^2$$

\* Calculate MR

$$MR = \frac{\Delta TR}{\Delta Q} = 45 - 1Q$$

$$= 45 - Q$$

\* assume that  $MC = 0.2Q$  what is the profit max output?

$$MR = MC$$

$$45 - Q = 0.2Q$$

$$45 = 1.2Q$$

$$Q^* = 37.5$$

\* Find the level of output that maximizes TR

$$TR \Rightarrow \max$$

$$MR \Rightarrow \text{zero}$$

$$45 - Q = 0$$

$$Q = 45 \rightarrow \max TR$$

